

Lab 6 : Navigation and Forms

basic_navigation

```
import 'package:flutter/material.dart';

void main() => runApp(const NavigationDemo());

class NavigationDemo extends StatelessWidget {
  const NavigationDemo({super.key});

  @override
  Widget build(BuildContext context) {
    return const MaterialApp(
      debugShowCheckedModeBanner: false,
      home: FirstScreen(),
    );
  }
}

class FirstScreen extends StatelessWidget {
  const FirstScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Login Page')),
      body: Center(
        child: FilledButton(
          onPressed: () {
            Navigator.push(
              context,
              MaterialPageRoute(builder: (_) => const SecondScreen()),
            );
          },
          child: const Text('Enter'),
        ),
      ),
    );
  }
}

class SecondScreen extends StatelessWidget {
  const SecondScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Welcome To Fitness Tracking App')),
      body: Center(
```

```
child: FilledButton(  
  onPressed: () => Navigator.pop(context),  
  child: const Text('Go Back'),  
),  
),  
);  
}  
}
```

Login Page

Enter

← Welcome To Fitness Tracking App

Go Back

pass_data

```
import 'package:flutter/material.dart';

void main() => runApp(const PassDataDemo());

class PassDataDemo extends StatelessWidget {
  const PassDataDemo({super.key});

  @override
  Widget build(BuildContext context) {
    return const MaterialApp(
      debugShowCheckedModeBanner: false,
      home: CourseListScreen(),
    );
  }
}

class CourseListScreen extends StatelessWidget {
  const CourseListScreen({super.key});

  @override
  Widget build(BuildContext context) {
    const courses = ['Sign up', 'Sign in', 'continue with Google', 'continue with Facebook', 'continue with microsoft'];

    return Scaffold(
      appBar: AppBar(title: const Text('Login Page')),
      body: ListView.builder(
        itemCount: courses.length,
        itemBuilder: (context, index) {
          final course = courses[index];

          return ListTile(
            title: Text(course),
            onTap: () {
              Navigator.push(
                context,
                MaterialPageRoute(
                  builder: (_) => CourseDetailsScreen(course: course),
                ),
              );
            },
          );
        },
      ),
    );
  }
}
```

```
class CourseDetailsScreen extends StatelessWidget {  
  const CourseDetailsScreen({required this.course, super.key});  
  
  final String course;  
  
  @override  
  Widget build(BuildContext context) {  
    return Scaffold(  
      appBar: AppBar(title: Text(course)),  
      body: Center(child: Text('Login through: $course')),  
    );  
  }  
}
```

Login Page

Sign up

Sign in

continue with Google

continue with Facebook

continue with microsoft

← continue with Google

Login through: continue with Google

← continue with microsoft

Login through: continue with microsoft

← Sign in

Login through: Sign in

form_validation

```
import 'package:flutter/material.dart';

void main() => runApp(const FormDemo());

class FormDemo extends StatelessWidget {
  const FormDemo({super.key});

  @override
  Widget build(BuildContext context) {
    return const MaterialApp(
      debugShowCheckedModeBanner: false,
      home: RegistrationScreen(),
    );
  }
}

class RegistrationScreen extends StatefulWidget {
  const RegistrationScreen({super.key});

  @override
  State<RegistrationScreen> createState() => _RegistrationScreenState();
}

class _RegistrationScreenState extends State<RegistrationScreen> {
  final formKey = GlobalKey<FormState>();
  final nameController = TextEditingController();
  final emailController = TextEditingController();

  @override
  void dispose() {
    nameController.dispose();
    emailController.dispose();
    super.dispose();
  }

  void submit() {
    if (formKey.currentState!.validate()) {
      ScaffoldMessenger.of(context).showSnackBar(
        SnackBar(
          content: Text('Welcome to Fitness tracker App'),
        ),
      );
    }
  }

  @override
  Widget build(BuildContext context) {
```

```

return Scaffold(
  appBar: AppBar(
    title: const Text('Registration Form'),
  ),

  body: Column(
    children: [
      const Padding(
        padding: EdgeInsets.all(16),
        child: Text(
          'Fitness Tracker App',
          style: TextStyle(
            fontSize: 24,
            fontWeight: FontWeight.bold,
          ),
        ),
      ),

      Expanded(
        child: Form(
          key: formKey,
          child: ListView(
            padding: const EdgeInsets.all(16),
            children: [
              TextFormField(
                controller: nameController,
                decoration: const InputDecoration(
                  labelText: 'Name',
                  border: OutlineInputBorder(),
                ),
                validator: (value) {
                  if (value == null || value.trim().isEmpty) {
                    return 'Name is required';
                  }
                  return null;
                },
              ),

              const SizedBox(height: 16),

              TextFormField(
                controller: emailController,
                keyboardType: TextInputType.emailAddress,
                decoration: const InputDecoration(
                  labelText: 'Email',
                  border: OutlineInputBorder(),
                ),
                validator: (value) {
                  if (value == null || !value.contains('@')) {

```

```
        return 'Enter a valid email';
      }
      return null;
    },
  ),

  const SizedBox(height: 20),

  FilledButton(
    onPressed: submit,
    child: const Text('Submit'),
  ),
],
),
),
),
),
],
),
);
}
}
```

Registration Form

Fitness Tracker App

Name	Khush
Email	khush@gmail.com
<input type="submit" value="Submit"/>	

Welcome to Fitness tracker App Khush

input_controls

```
import 'package:flutter/material.dart';

void main() => runApp(const InputControlsDemo());

class InputControlsDemo extends StatefulWidget {
  const InputControlsDemo({super.key});

  @override
  State<InputControlsDemo> createState() => _InputControlsDemoState();
}

class _InputControlsDemoState extends State<InputControlsDemo> {
  String priority = 'Medium';
  bool reminder = true;
  bool completed = false;

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: Scaffold(
        appBar: AppBar(title: const Text('Fitness Tracker App')),
        body: ListView(
          padding: const EdgeInsets.all(16),
          children: [
            DropdownButtonFormField<String>(
              initialValue: priority,
              decoration: const InputDecoration(
                labelText: 'Priority',
                border: OutlineInputBorder(),
              ),
              items: ['Low', 'Medium', 'High']
                .map((value) => DropdownMenuItem(
                  value: value,
                  child: Text(value),
                ))
                .toList(),
              onChanged: (value) {
                setState(() => priority = value!);
              },
            ),
            SwitchListTile(
              title: const Text('Enable Notification reminder'),
              value: reminder,
              onChanged: (value) {
                setState(() => reminder = value);
              },
            ),
          ],
        ),
      ),
    );
  }
}
```

```
CheckboxListTile(  
  title: const Text('Terms and Conditions'),  
  value: completed,  
  onChanged: (value) {  
    setState(() => completed = value ?? false);  
  },  
),  
],  
),  
),  
);  
}  
}
```

Fitness Tracker App

Priority

High

Enable Notification reminder

☐

Terms and Conditions

☒

dialog_bottom_sheet

```
import 'package:flutter/material.dart';

void main() => runApp(const FeedbackDemo());

class FeedbackDemo extends StatelessWidget {
  const FeedbackDemo({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: Scaffold(
        appBar: AppBar(title: const Text('Fitness Tracker App')),
        body: Builder(
          builder: (context) => Center(
            child: Column(
              mainAxisAlignment: MainAxisAlignment.min,
              children: [
                FilledButton(
                  onPressed: () => showDialog<void>(
                    context: context,
                    builder: (_) => AlertDialog(
                      title: const Text('Confirm'),
                      content: const Text('Do you want to continue?'),
                      actions: [
                        TextButton(
                          onPressed: () => Navigator.pop(context),
                          child: const Text('Cancel'),
                        ),
                        FilledButton(
                          onPressed: () => Navigator.pop(context),
                          child: const Text('Continue'),
                        ),
                      ],
                    ),
                  ),
                const Text('Permission'),
                const SizedBox(height: 12),
                OutlinedButton(
                  onPressed: () => showModalBottomSheet<void>(
                    context: context,
                    builder: (_) => const Padding(
                      padding: EdgeInsets.all(24),
                      child: Text('This is a Notification.'),
                    ),
                ),
                const Text('Notification'),
              ],
            ),
          ),
        ),
      ),
    );
  }
}
```

```
        ),
    ],
    ),
    ),
    ),
    ),
    );
}
```

Fitness Tracker App

Permission

Notification

Fitness Tracker App

Confirm

Do you want to continue?

Cancel

Continue

Permission

Notification

This is a Notification.

```

import 'package:flutter/material.dart';
import '../models/task.dart';
import 'add_task_screen.dart';
import 'task_details_screen.dart';

class TaskListScreen extends StatefulWidget {
  const TaskListScreen({super.key});

  @override
  State<TaskListScreen> createState() => _TaskListScreenState();
}

class _TaskListScreenState extends State<TaskListScreen> {
  final tasks = <Task>[
    Task(
      title: 'Complete Flutter UI',
      subject: 'SDP',
      priority: 'High',
      description: 'Finish the Lab 5 interface.',
    ),
  ];

  Future<void> openAddTask() async {
    final task = await Navigator.push<Task>(
      context,
      MaterialPageRoute(builder: (_) => const AddTaskScreen()),
    );

    if (task != null) {
      setState(() => tasks.add(task));
    }
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Student Task Manager')),
      body: ListView.builder(
        itemCount: tasks.length,
        itemBuilder: (context, index) {
          final task = tasks[index];

          return Card(
            child: ListTile(
              title: Text(task.title),
              subtitle: Text('${task.subject} - ${task.priority} priority'),
              trailing: const Icon(Icons.chevron_right),
              onTap: () {
                Navigator.push(

```

```

        context,
        MaterialPageRoute(
          builder: (_) => TaskDetailsScreen(task: task),
        ),
      ),
    },
  ),
);
},
),
floatingActionButton: FloatingActionButton.extended(
  onPressed: openAddTask,
  icon: const Icon(Icons.add),
  label: const Text('Add Task'),
),
);
}
}

```

Student Task Manager

Complete Flutter UI
SDP - High priority



+ Add Task